Supplemental Materials for 'Temporal Generalized Linear Mixed Models to Monitor National Health Behaviour Using Subnational Observational Data'

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Sources of Data

National and State-Specific Surveys of Adolescent E-Cigarette Use Prevalence

Survey Name	Source/Provider	Years Surveyed [a]	Grades Surveyed [b]	Questionnaire/Codebook/Report/Website Wording
National				
National Youth Tobacco Survey	US Centers for Disease Control and Prevention (2025); (US Food and Drug Administation, 2026)	2011–2025	High School	“During the past 30 days, which of the following products/tobacco products did you use/have you used on at least one day?... Electronic Cigarettes or E-cigarettes, such as Ruyan or NJOY” (2011–2013); “During the past 30 days, on how many days did you use electronic cigarettes or e-cigarettes such as blu, 21st Century Smoke, or NJOY?” (2014); “During the past 30 days, on how many days did you use electronic cigarettes or e-cigarettes?” (2015); “During the past 30 days, on how many days did you use e-cigarettes?” (2016–2025)
State-Level (Alphabetical by State)				
Arizona Youth Survey (AYS)	Arizona Criminal Justice Commission (2026)	2016–2024	10, 12	"During the past 30 days, on how many days did you use electronic cigarettes (e-cigarettes)?" (2016); "During the past 30 days, on how many occasions, if any, have you... used electronic cigarettes (e.g., e-cigs, vapes)?" (2018); "During the past 30 days, have you... vaped e-liquid or e-juice with nicotine (e.g., e-cigarettes)?" (2020–2022); "During the past 30 days, on how many DAYS (if any) have you: vaped e-juice/e-liquid with nicotine (e.g., e-cigarettes)?" (2024)
Arkansas Prevention Needs Assessment Survey (APNA)	Arkansas Department of Human Services and International Survey Associates LLC (2026)	2020–2024	10, 12	"On how many occasions (if any) have you: vaped NICOTINE during the past 30 days?" (2020–2024)
California Student Tobacco Survey (CSTS); California Youth Tobacco Survey (CYTS)	University of California San Diego (2021); California Department of Public Health (2026)	2016–2020; 2022–2024	High School	“Current use of… E-cigarettes” (2016–2018); “Current use of… Vapes” (2020–2024)
Healthy Kids Colorado Survey (HKCS)	Colorado Department of Public Health & Environment (2026)	2015–2023	High School	"Percentage of students who used an electronic vapor product in the past 30 days" (2015–2023)
Delaware School Survey (DSS)	University of Delaware (2026)	2014–2024	11	"Delaware students’ self reported e-cigarette and/or vaping device use in the past 30 days… 2014-2015: Only e-cigarettes; 2016: E-cigarettes and vaping asked about in the same question; 2017: E-cigarettes and vaping asked about in separate questions, both reported here; 2018-2023: Specific brands also asked about (e.g., Juul)"
Florida Youth Tobacco Survey (FYTS)	Florida Department of Health (2025)	2011–2024	High School	"The wording of survey questions asking about electronic vaping has changed over time. From 2011 to 2014, the survey asked students about use of electronic cigarettes. In 2015 and later years, the survey asked about use of electronic vapor products."

Survey Name	Source/Provider	Years Surveyed [a]	Grades Surveyed [b]	Questionnaire/Codebook/Report/Website Wording
Georgia Student Health Survey (GSHS) Georgia Student Wellness Survey (GSWS)	Georgia Department of Education (2026)	2015–2020, 2022–2024; 2021	9, 10, 11, 12	"During the past 30 days, on how many days did you smoke an electronic vapor product?" (2015–2021)"; "During the past 30 days, on how many days did you smoke an electronic vapor product (such as Juul, e-cigars, e-pipes, vape pipes, vaping pens, e-hookahs, or hookah pens)?" (2022–2024)
Hawaii Youth Tobacco Survey (HYTS)	Hawaii State Department of Health (2026)	2017–2019	High School	"During the past 30 days, on how many days did you vape (use an electronic vapor product)?" (2019); "During the past 30 days, on how many days did you use e-cigarettes?" (2017)
Idaho Healthy Youth Survey (IHYS)	Idaho Office of Drug Policy (2026)	2017–2021	10, 12	"During the past 30 days, on how many days did you use a vape pen or e-cigarette?" (2017–2021)
Illinois Youth Survey (IYS)	University of Illinois (2026)	2016–2024	10, 12	“E-cigarettes in the past 30 days” (2016–2018); "e-cigarettes or other vaping products during the past 30 days" (2024)
Indiana Youth Survey (IYS)	Indiana University Bloomington (2026)	2015–2024	9, 10, 11, 12	"How many times in the last month (30 days) have you used … ? Electronic vapor products (e-cigarettes, vaping pens, e-hookahs, etc.)" (2015–2024)
Iowa Youth Survey (IYS)	Iowa Department of Health and Human Services (2026)	2014–2021	11	"In the past 30 days, on how many days have you used electronic cigarettes or e-cigarettes (battery-powered cigarettes)?" (2014); "In the past 30 days, on how many days have you used e-cigarettes (vape-pens, hookah-pens, e-hookahs, e-cigars, e-pipes, personal vaporizers or mods)?" (2016); "In the past 30 days, on how many days have you used e-cigarettes (vape-pens, JUUL, hookah-pens, e-hookahs, e-cigars, e-pipes, personal vaporizers or mods)?" (2018); "Used an e-cigarette, JUUL, vape-pen, ehookah, mod-box, or other electronic cigarette" (2021)
Kansas Communities That Care Student Survey (KCTC)	Kansas Department for Aging and Disability Services (2025)	2016–2025	10, 12	"During the past 30 days, on how many days have you used electronic cigarettes (e-cigarettes)?" (2016–2020); "How frequently have you vaped during the past 30 days?" (2021–2025)
Kentucky Incentives for Prevention (KIP)	Kentucky Cabinet for Health and Family Services (2026)	2014–2024	10	“Electronic Cigarettes or E-cigarettes,* 30-Day Use… *for ex. NJOY, V2, Bull Smoke, Halo” (2014–2016); "During the past 30 days, which of the following tobacco products did you use on at least one day? (You can choose ONE ANSWER or MORE THAN ONE ANSWER.)... Vape pens or e-cigarettes (for ex., NJOY, V2, Bull Smoke, Halo or JUUL)" (2018); "The next 9 questions ask about electronic vapor products, such as JUUL, Vuse, MarkTen, Alto, Puff and blu. Electronic vapor products include e-cigarettes, vapes, vape pens, e-cigars, e-hookahs, hookah pens, and mods. Do not count any electronic vapor products used for smoking marijuana when answering these questions... On how many occasions (if any) have you vaped/used an electronic vapor product... in the past 30 days?" (2021); "On how many occasions (if any) have you vaped/used an electronic vapor product in the past 30 days?" (2024)
Louisiana Youth Tobacco Survey (LYTS)	(Louisiana Department of Health, 2026)	2015–2023	High School	“E-Cigarettes” (2015–2017); “Vape” (2017–2023)

Survey Name	Source/Provider	Years Surveyed [a]	Grades Surveyed [b]	Questionnaire/Codebook/Report/Website Wording
Maine Integrated Youth Health Survey (IYHS)	Maine Department of Health and Human Services (2025)	2015–2025	High School	"During the past 30 days, on how many days did you use an electronic vapor product?" (2015–2025)
Massachusetts FC/NQ Student Health Survey (SHS)	Communities That Care Coalition (2025)	2016–2025	8, 10, 12 (pre-aggregated; could not exclude 8)	"On how many occasions (if any) have you: used electronic cigarettes, e-cigarettes, vape pens, or e-hookahs in your lifetime (or during the past 30 days)?" (2018, 2022, 2025); "Past 30 Days… Vaping" (Other years)
Minnesota Student Survey (MSS)	Minnesota Department of Education (2025)	2016–2025	9, 11	"During the last 30 days, on how many days did you... use an electronic cigarette (ecigarette, e-hookah, vaping pen)?" (2016); "During the last 30 days, on how many days did you… vape or use an e-cigarette like JUUL, suorin, blu, VUSE, or logic?" (2019); "During the last 30 days, on how many days did you… vape or use an e-cigarette that contains nicotine, such as JUUL, VUSE, NJOY, Puff Bar, Blu, or Bidi Stick?" (2022–2025)
Mississippi Youth Tobacco Survey (YTS)	Mississippi State Department of Health (2025)	2011–2025	High School	"Current E-Cig" (2011–2025); "Changes between 2013 and 2014 in the wording and placement of questions about the use of e-cigarettes might have had an impact on reported use of these products"
Missouri Student Survey (MSS)	Missouri Department of Mental Health (2026)	2014–2024	6, 9, 12 (pre-aggregated; could not exclude 6)	"What is your best estimate of the number of days you used electronic cigarettes (e-cigs) during the past 30 days?" (2014); "What is your best estimate of the number of days you used e-cigs, mods or vapes during the past 30 days?" (2016–2022); "What is your best estimate of the number of days you used vape/vaping devices during the past 30 days?" (2024)
Montana Prevention Needs Assessment (MPNA)	Montana Department of Public Health & Human Services (2026)	2016–2024	10, 12	"In the past 30 days, on how many occasions (if any) have you… used electronic cigarettes, e-cigarettes, vape pens, or e-hookahs?" (2016–2024)
Nebraska SHARP Youth Tobacco Survey (NSYTS)	Nebraska Bureau of Sociological Research (2026)	2013–2023	High School	"Current e-cigarette use was defined as using electronic cigarette or e-cigarette on at least one day during the 30 days preceding the survey" (2013); "Current e-cigarette use was defined as using an electronic cigarette or e-cigarette such as Ruyan or NJOY on at least one day during the 30 days preceding the survey" (2015–2017); "Electronic Cigarettes or e-cigarettes, such as Ruyan, NJOY, or JUUL" (2019–2022); "During the PAST 30 DAYS, on how many days did you use e-cigarettes?" (2023)
New Jersey Youth Tobacco Survey (NJYTS)	New Jersey Department of Health (2026)	2012–2022	High School	"Current E-Cigarette Use" (2012–2014); "Current (Past 30-days) Use of… E-cig: Electronic cigarette" (2016–2022)
New York Youth Tobacco Survey (NYYTS)	New York State Department of Health (2026)	2014–2024	High School	"During the past 30 days, on how many days did you use electronic cigarettes or e-cigarettes such as blu, 21st Century Smoke, or NJOY?" (2014); "During the past 30 days, on how many days did you use an e-cigarette, vape pen, hookah pen, or e-hookah?" (2016–2018); "During the past 30 days, on how many days did you vape?" (2020–2024)
North Carolina Youth Tobacco Survey (NCYTS)	North Carolina Department of Health and Human Services (2026)	2011–2022	High School	"In the past 30 days, which of the following products have you used on at least one day?... Electronic Cigarettes or E-cigarettes, such as Ruyan or NJOY" (2011–2013); "In the past 30 days, which of the following products have you used on at least one day?... Electronic Cigarettes or E-cigarettes, also known as e-hookah, hookah pens, or vape pens, such as Vuse, blu, Fin, Logic, NJOY, and e-Go" (2015); "During the past 30 days, on how many days did you use e-cigarettes?" (2017–2022)

Survey Name	Source/Provider	Years Surveyed [a]	Grades Surveyed [b]	Questionnaire/Codebook/Report/Website Wording
Ohio Healthy Youth Environments Survey (OHYES!)	Ohio Department of Mental Health & Addiction Services et al. (2026)	2016–2024	High School, Middle School (pre-aggregated; could not exclude Middle School)	"During the past 30 days, on how many days did you use an electronic vapor product?" (2016–2024)
Oklahoma Prevention Needs Assessment Survey (OPNA)	Oklahoma State Department of Mental Health and Substance Abuse (2026)	2022–2024	10, 12	"Past 30 Day Vaping Product or E-Cigarette Use (For Nicotine or Tobacco Purposes)" (2022–2024)
Oregon Healthy Teens Survey (OHTS); Oregon Student Health Survey (OSHS)	Oregon Health Authority (2026a); Oregon Health Authority (2026b)	2013-2019; 2020–2024	11	"During the past 30 days, on how many days did you smoke an electronic nicotine delivery product, such as an e-cigarette, e-cigar, or e-hookah?" (2013); "During the past 30 days, on how many days did you use an e-cigarette or other vaping product?" (2015–2019); "During the past 30 days, on how many days... Did you use e-cigarettes or other vaping products, such as Juul?" (2020–2022); "During the past 30 days, which products have you used?... Vaping product or other e-cigarettes" (2024)
Pennsylvania Youth Survey (PAYS)	Commonwealth of Pennsylvania Commission on Crime and Delinquency (2026)	2015–2023	10, 12	"How frequently have you used an electronic vapor product such as e-cigarettes, e-cigars, e-pipes, vape pipes, vaping pens, e-hookahs, or hookah pens during the past 30 days?" (2015–2017); "How frequently have you used an electronic vapor product such as e-cigarettes, e-cigars, e-pipes, vape pens, e-hookahs, or hookah pens during the past 30 days?" (2019); "How frequently have you used an electronic vapor product such as: JUUL, Vuse, MarkTen, and blu or other e-cigarettes vapes vape pens e-cigars, e-hookahs, hookah pens and mods during the past 30 days?" (2021); "How often have you, In the past 30 days: Used an electronic vapor product such as e-cigarettes, vapes, mods, e-cigs, e-hookahs, or vape pens?" (2023)
Rhode Island Student Survey (RISS)	State of Rhode Island State Epidemiological Outcomes Workgroup (2026)	2018–2024	High School	"Electronic Nicotine Delivery Systems (ENDS), also known as e-cigarettes or vape pens or devices" (2018–2024)
South Carolina Youth Tobacco Survey (SCYTS)	South Carolina Department of Public Health (2025)	2017–2024	High School	"e-Cigarette" (2017-2024)
South Dakota Youth Tobacco Survey (SDYTS)	South Dakota Department of Health (2026)	2011–2023	Middle School (Not conducted for High School)	“Electronic cigarettes” (2011); “Electronic cigarettes, such as Ruyan or NJoy” (2013); “E-cigarettes” (2015–2023)
Tennessee Together Student Survey (TTSS)	Tennessee Department of Mental Health & Substance Abuse Services (2026)	2019–2023	10, 12	"During the past 30 days, on how many days did you use an e-cigarette, e-cigar, or e-hookah?" (2019); "During the past 30 days, on how many days did you use a vaping device with nicotine?" (2023)

Survey Name	Source/Provider	Years Surveyed [a]	Grades Surveyed [b]	Questionnaire/Codebook/Report/Website Wording
Texas Youth Tobacco Survey (TYTS)	Texas Department of State Health Services (2026)	2014–2025	High School	“During the past 30 days, on how many days did you use electronic cigarettes or e-cigarettes, such as Ruyan or NJOY?” (2014); “During the past 30 days, on how many days did you use electronic cigarettes, also called e-cigarettes, vape pens, e-hookah, hookah pens, and e-cigars, such as NJOY, blu, or Logic?” (2016); “Electronic Cigarettes or E-Cigarettes” (2018–2025)
Utah Student Health and Risk Prevention (SHARP) Survey	Utah Department of Health and Human Services (2025)	2019–2025	10, 12	“vape products such as e-cigarettes, vape pens, mods, or pod vapes like JUUL or Puff Bars” (2019–2025)
Washington State Healthy Youth Survey (WSHYS)	Washington State Health Care Authority et al. (2026)	2012–2023	10, 12	"During the past 30 days, on how many days did you... Use electronic cigarettes or e-cigs?" (2012); "During the past 30 days, on how many days did you... Use an electronic cigarette, also called e-cigs, or vape pens?" (2014–2018); "During the past 30 days, on how many days did you... Use an electronic cigarette, also called e-cigs, JUUL, or vape pens?" (2021); "During the past 30 days, on how many days did you use an electronic cigarette, also called e-cigs, device pens, or vape pens?" (2023)
West Virginia Youth Tobacco Survey (WVYTS)	West Virginia Department of Health & Human Resources (2026); West Virginia Department of Health & Human Resources (2020)	2013–2015	High School	"Current Use of an Electronic Vapor Product" (2013–2015)
Wisconsin Youth Tobacco Survey (WYTS)	Wisconsin Department of Health Services (2023); Palmersheim (2019)	2012–2022	High School	“E-Cigarettes” (2012–2018); “E-cigarettes (vapes)” (2022)
Wyoming Prevention Needs Assessment (WPNA)	Wyoming Department of Health (2026)	2018–2024	10, 12	“On how many occasions (if any) have you vaped during the past 30 days?” (2018); “During the past 30 days, how many days did you juul, vape or use an electronic vapor product?” (2020–2024)

[a] With e-cigarette use measures.

[b] As included in this work. Other grades that were surveyed were not included in this work.

Mathematical Description of Models

Mathematically, the model is specified by the expectation for the ith Year and j^{th} State

$$\mathrm{E}(\text{National Prev}_{i,j}) = \text{logit}^{-1}\left(\beta_0 + \beta_1 \text{State Prev}_{i,j} + \beta_{2,j}\text{State}_j + \beta_{3,j}\text{State Prev}_{i,j}\text{State}_j\right)$$

(note that the linear predictor contains only fixed effects so the model is marginal) and the variance for the jth State

$$\text{Var}(\text{National Prev}_j) = \varphi \boldsymbol{A}_j^{\frac{1}{2}} \boldsymbol{R}_j \boldsymbol{A}_j^{\frac{1}{2}}$$

where φ is the scale, $\boldsymbol{R}$ is the SP(POW) matrix, and $\boldsymbol{A}$ is the variance function matrix determined by the choice of response distribution.

Weighting Procedure

SAS accounts for the magnitude of events and counts when fitting the model, which can cause issues when the survey-weighted events/counts are orders of magnitude larger than the unweighted events/counts. To address this, the survey-weighted events/counts were down-weighted such that the down-weighted events divided by the down-weighted trials still produces the same weighted national prevalence estimate each year as the weighted events divided by the weighted trials (to be nationally representative), but the number of (down-weighted) trials the model 'sees' equals the *unweighted* NYTS sample size. More completely, the survey-weighted events and survey-weighted trials from NYTS were down-weighted so that:

(1) For a given year, the sum of the down-weighted trials values across states equals the unweighted sample size of NYTS that year, with the value of down-weighted trials for each state being proportional to the population of that state in that year (taken from the United States Census Bureau (US Census Bureau, 2021, 2026)). As an illustrative example, if the only states with 2014 data were Mississippi and Texas, the populations of these states in 2014 were 2,990,468 and 26,964,333 respectively, and the unweighted NYTS sample size in 2014 was 22,007, then the value of the down-weighted trials for Mississippi in 2014 would be 22,007×[2,990,468÷(2,990,468+26,964,333)]≈2,197, and the value of the down-

weighted trials for Texas in 2014 would be 22,007×[26,964,333÷(2,990,468+26,964,333)] ≈19,810. The sum of these recovers the unweighted NYTS sample size, i.e. 2,197+19,810=22,007 as required.

(2) For a given year, the down-weighted events value divided by the down-weighted trials value for each state equals the weighted NYTS prevalence that year, and the sum of the down-weighted events values across states divided by the sum of the down-weighted trials values across states also equals the weighted NYTS prevalence that year. Continuing the example above, if the weighted NYTS prevalence in 2014 was 13.4% or 0.134, then the value of the down-weighted events for Mississippi in 2014 would be 0.134×2,197≈294, and the value of the down-weighted events for Texas in 2014 would be 0.134×19,810≈2,655. The weighted NYTS prevalence is then recovered as 294÷2,197≈0.134 for Mississippi, 2,655÷19,810≈0.134 for Texas, and (294+2,655)÷(2,197+19,810)≈0.134 across the two states as required.

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